ELSEVIER

Journal of Econometrics 104 (2001) 179–207

JOURNAL OF Econometrics

www.elsevier.com/locate/econbase

A nonlinear autoregressive conditional duration model with applications to financial transaction data

Michael Yuanjie Zhang[a], Jeffrey R. Russell[b,*], Ruey S. Tsay[b]

[a] *Barr Rosenberg Research Center, AXA Rosenberg Group, 4 Orinda Way Bldg E, Orinda, CA 94563, USA*

[b] *Graduate School of Business, University of Chicago, 1101 East 58th Street, Chicago, IL 60637, USA*

Received 4 October 1999; revised 15 November 2000; accepted 23 January 2001

Abstract

This paper presents a new model that improves upon several inadequacies of the original autoregressive conditional duration (ACD) model considered in Engle and Russell (Econometrica 66(5) (1998) 1127–1162). We propose a threshold autoregressive conditional duration (TACD) model to allow the expected duration to depend nonlinearly on past information variables. Conditions for the TACD process to be ergodic and existence of moments are established. Strong evidence is provided to suggest that fast transacting periods and slow transacting periods of NYSE stocks have quite different dynamics. Based on the improved model, we identify multiple structural breaks in the transaction duration data considered, and those break points match nicely with real economic events.

JEL classification: C22; C41; G14

Keywords: Nonlinear time series; Autoregressive conditional duration; Structural break; Duration models; Market microstructure

* Corresponding author. Tel.: +1-773-834-0720; fax: +1-773-703-0458.
E-mail addresses: mzhang@axarosenberg.com (M.Y. Zhang), jeffrey.russell@gsb.uchicago.edu (J.R. Russell), ruey.tsay@gsb.uchicago.edu (R.S. Tsay).

0304-4076/01/$ - see front matter
PII: S0304-4076(01)00063-X

1. Introduction

High-frequency financial transaction data provide researchers with unprecedented opportunities to study the evolution of financial asset prices at the transaction by transaction frequency. These data sets have generated enormous interest including testing of market microstructure theories and obtaining more precise estimates of volatility. [1] These analysis are complicated by the fact that the data arrive in unequally spaced time intervals. Engle and Russell (1998), henceforth ER, propose a new model for the arrival times of trades, [2] termed the autoregressive conditional duration (ACD) model. They find the ACD model provides a good description for the IBM data analyzed. However, there are still features of the data such as nonlinearities and excess dispersions that cannot be fully captured by their linear ACD model with Exponential or Weibull errors.

The objective of this paper is to construct models that better describe the duration between trades of financial assets and shed light on the dynamics of the transaction arrival process. We propose a threshold ACD model which allows the conditional expected duration to be a nonlinear function of the past information variables. Characterizations of geometric ergodicity and existence of moments for the proposed model are investigated. Since ER find that the Exponential and Weibull specifications suffer from remaining excess dispersion, we use generalized Gamma distribution in the threshold ACD framework and allow the shape parameters to be different across regimes. We relate the different regimes to other observable market characteristics including bid-ask spreads, volume, and volatility. Consistent with market microstructure theories, we find evidence that fast transacting regimes are associated with agents trading based on private information.

We also find evidence that nonstationarity may be an important feature of the transaction durations. Even with the added flexibility of our proposed nonlinear model we still identify multiple structural breaks for the transaction duration series considered. After accounting for the structural breaks, the proposed threshold ACD model provides a good characterization of the transaction arrival behavior of each subperiod.

[1] See for example the HFDF I and II proceedings (Olsen and Associates, 1995, 1998) for various applications of high-frequency data and Andersen and Bollerslev (1998) and Bai et al. (1999) for discussion of estimating volatility using high-frequency data. Also see Goodhart and O'Hara (1997) for an extensive review of applications of market microstructure theory on high-frequency financial data.

[2] We use the term 'transaction arrival times' and 'transaction durations' inter-changeably throughout the text, since, from the point of view of a point process, it is equivalent to model (1) transaction arrival times, (2) transaction durations—time intervals between consecutive trades, or (3) the counting measure for the trade arrival process.

The rest of the paper is organized as follows. In Section 2, we propose a threshold ACD model and investigate some properties of the model. Section 3 provides a detailed description of our modeling approach for the IBM transaction duration data using a 3-regime threshold ACD model. Section 4 discusses the economic and statistical significance of the structural breaks we detected. Section 5 explores the economic meanings of regimes and the implications of the threshold dynamics. Section 6 concludes.

2. A threshold ACD model

ER find that simple versions of the ACD model do a good job of capturing the temporal dependence in the IBM data analyzed, but model diagnostics suggest several directions in which the model can be improved. First, model diagnostics provide evidence of nonlinearity in the conditional mean. In this section, we propose a nonlinear ACD-type model. Threshold models can be viewed as a simple nonlinear form to approximate a complex stochastic system by introducing simpler subsystems through a threshold variable (Tong, 1990). It is well known that the threshold autoregressive (TAR) model is capable of producing features such as asymmetric periodic behavior and jump phenomena. Among others, systematic modeling procedures for TAR models can be found in Tsay (1989) and Tsay (1998) for univariate case and multivariate case, respectively. The proposed threshold ACD model is closely related to the TAR model and the more general threshold autoregressive moving average (ARMA) model.

2.1. The ACD framework

Let t_i be the ith transaction time where $0 = t_0 < t_1 < \cdots < t_N$, and $X_i = t_i - t_{i-1}$ be the duration between trades. The basic assumption of the ACD model is

$$X_i/\psi_i \equiv \varepsilon_i \sim \text{ i.i.d. } \mathscr{D}(\theta_2), \tag{1}$$

where $\psi_i \equiv \mathbb{E}[\mathbf{X}_i|\mathbf{X}_{i-1},\ldots,\mathbf{X}_1;\theta_1]$ is the conditional mean duration of the ith trade with parameter vector θ_1 and $\mathscr{D}$ is a general distribution over $(0,\infty)$ with mean equal to one and parameter vector θ_2. The ACD model assumes that the error term is multiplicative and all past information enters the current duration through the conditional mean duration ψ_i. The flexibility of the model lies in the rich host of candidates for the distribution $\mathscr{D}$ as well as the specification of the dynamic structure of the conditional mean duration ψ_i. ER use both Exponential and Weibull distributions in their analysis of an IBM's series.

An ACD(p, q) model specifies the conditional mean duration ψ_i as a linear function of lagged durations and their conditional expectations

$$\psi_i = \alpha_0 + \sum_{m=1}^{p} \alpha_m X_{i-m} + \sum_{n=1}^{q} \beta_n \psi_{i-n}. \tag{2}$$

The model is closely related to the GARCH-type model of Engle (1982) and Bollerslev (1986). Forecasts of the model can be obtained using the standard ARMA approach, since, like GARCH models, this model also has an ARMA representation.

2.2. The threshold ACD

The threshold ACD model is a simple but powerful generalization of the ACD model that allows subregimes to have different persistence, conditional means and error distributions. These differentials can potentially generate a rich host of nonlinear dynamics. We first spell out the general threshold ACD framework and then define the threshold ACD(1,1) model used in our empirical modeling of the transaction durations.

Let $\{X_i\}$ be a positive stochastic process and $\psi_i = \mathbb{E}[X_i|\mathscr{F}_{i-1};\theta_1]$ be the conditional mean of X_i. Define $R_j = [r_{j-1}, r_j)$, $j = 1, 2, \ldots, J$, for a positive integer J, where $-\infty = r_0 < r_1 < \cdots < r_J = \infty$ are the threshold values. The process $\{X_i\}$ follows a J-regime threshold ACD model if, when the threshold variable $Z_{i-d} \in R_j$,

$$X_i/\psi_i \equiv \varepsilon_i \sim \text{ independent } \mathscr{D}(\theta_2^{(j)}), \tag{3}$$

$$\psi_i = \alpha_0^{(j)} + \sum_{m=1}^{p} \alpha_m^{(j)} X_{i-m} + \sum_{n=1}^{q} \beta_n^{(j)} \psi_{i-n}, \tag{4}$$

where the delay parameter d is a positive integer, and $\mathscr{D}$ is a general distribution with support $(0, \infty)$ and mean equal to one. The general form of the threshold variable $\{Z_i\}$ is given by $Z_i = h(X_i, \ldots, X_1; Y_i, \ldots, Y_1)$, where $\{Y_i\}$ is a vector of economic variables.

ER discuss the exact relationship between the distribution of the error term and the associated hazard function that is often of interest in duration models. Here we allow the parameter θ_2 of the error distribution $\mathscr{D}$ to be different across regimes to allow for distinct shapes in hazard function in different trading regimes. We can also use exogenous variables as additional regressors in Eq. (4).

As in the ARCH literature, it is informative to write the threshold ACD model in a threshold ARMA form. Let $\eta_i = X_i - \psi_i$, $\alpha_m^{(j)} = 0$ for $m = p + 1, \ldots, \max(p, q)$, and $\beta_n^{(j)} = 0$ for $n = q + 1, \ldots, \max(p, q)$, for $j = 1, \ldots, J$.

The model can be written as, if $Z_{i-d} \in R_j$,

$$X_i = \alpha_0^{(j)} + \sum_{m=1}^{\max(p,q)} (\alpha_m^{(j)} + \beta_m^{(j)}) X_{i-m} + \eta_i - \sum_{n=1}^{q} \beta_n^{(j)} \eta_{i-n}, \tag{5}$$

which is a threshold ARMA$(J; \max(p,q), q)$ process, but with conditional heteroskedastic innovations.

Zakoian (1994) and Rabemananjara and Zakoian (1993) propose a threshold GARCH model to capture the asymmetries in volatility, with 'zero' as the threshold value for zero mean innovation. The proposed model can be considered as a generalization of their model but with very different characteristics. Glosten et al. (1993) present an important application of threshold GARCH model in asset pricing settings.

Just as the ACD(1,1) and GARCH(1,1) models provide natural starting points so does the Threshold ACD(1,1) model with lag-1 response as the threshold variable. We will refer this model as TACD(1,1) model throughout the paper.

Definition 1. A positive sequence $\{X_i\}$ follows a TACD(1,1) process if

$$\begin{cases} X_i = \psi_i \varepsilon_i^{(j)} \\ \psi_i = \alpha_0^{(j)} + \alpha_1^{(j)} X_{i-1} + \beta_1^{(j)} \psi_{i-1} \end{cases} \quad \text{if } X_{i-1} \in R_j, \tag{6}$$

where $R_j = [r_{j-1}, r_j)$, $j = 1, 2, \ldots, J$, where the positive integer J is the number of regimes and $0 = r_0 < r_1 < \cdots < r_J = \infty$ are the threshold values. $\alpha_0^{(j)} > 0$, $\alpha_1^{(j)} \geqslant 0$, and $\beta_1^{(j)} \geqslant 0$ for $j = 1, 2, \ldots, J$. For fixed j, $\{\varepsilon_i^{(j)}\}$ is an iid sequence with positive density $f^{(j)}(\cdot)$ over $(0, \infty)$ and $\mathbb{E}[\varepsilon_i^{(j)}] = 1$. $\{\varepsilon_i^{(j)}\}$ and $\{\varepsilon_i^{(k)}\}$ are independent for $j \neq k$.

Section 3 applies the TACD(1,1) model to IBM transactions data. We now establish some properties of the TACD(1,1) model.

2.3. Geometric ergodicity and existence of moments of TACD(1,1)

This section provides sufficient conditions for geometric ergodicity and existence of moments of the stationary distribution for the TACD(1,1) model in two theorems. Our first theorem gives a sufficient condition for geometric ergodicity of the model. Tong (1990) provides a general framework using drift criteria to characterize nonlinear time series, and Meyn and Tweedie (1993) provide a rigorous treatment of drift criteria and other methods in a general Markov setting. Petruccelli and Woolford (1984), Chan et al. (1985) and Chen and Tsay (1991) obtain necessary and sufficient conditions for the geometric ergodicity of a general TAR(1) model, but it is difficult to derive a necessary and sufficient condition for a given class of nonlinear time series models in general. Brockwell et al. (1992) give a sufficient condition of strict stationarity for a subclass of threshold ARMA model, namely, threshold ARMA models with continuity at regime boundaries and same moving average coefficients across regimes. The generalized autoregressive approach used by Bougerol and Picard (1992) to prove the stationarity and ergodicity of GARCH(p, q) process, is not applicable here because of the nonlinear feature of our model. Proofs of the theorems are given in an appendix.

Theorem 1. A TACD$(1,1)$ *process* $\{X_i\}$ *is geometrically ergodic if*

$$\text{(i)} \quad \alpha_1^{(j)} > 0, \quad \beta_1^{(j)} > 0 \quad \textit{for } j = 1,\ldots,J, \tag{7}$$

$$\text{(ii)} \quad \alpha_1^{(J)} + \beta_1^{(J)} < 1. \tag{8}$$

The sufficient condition (i) of the theorem is necessary in proving the irreducibility of the process. The sufficient condition (ii) of the theorem is easy to understand. Since the process is bounded below by zero, the only chance that the process can 'blow up' is through regime J. The condition that the autoregressive coefficient in regime J, $\alpha_1^{(J)} + \beta_1^{(J)}$, is smaller than one will regulate the process to be well behaved once it reached regime J. It is important to note that the process is stable even if it has exploding or unit root behavior in regime 1 through regime $J - 1$.

Using geometric ergodicity, we can evaluate the stationary distribution and other related joint and conditional distributions numerically, though analytical results are hard to obtain for nonlinear models. For a geometric ergodic process, Moeanaddin and Tong (1990) propose a conditional density approach by exploiting the Chapman–Kolmogorov relation.

Our next theorem provides a sufficient condition for the existence of moments of TACD$(1,1)$ process.

Theorem 2. For a given integer $k \geqslant 1$, *assume* $\mathbb{E}[(\varepsilon_i^{(j)})^k] < \infty$, *for* $j = 1,\ldots,J$. *Then, the invariant probability measure of the TACD*$(1,1)$ *process given in Theorem* 1 *has finite moments up to order k if*

$$\text{(i)} \quad \alpha_1^{(j)} > 0, \quad \beta_1^{(j)} > 0, \textit{ for } j = 1,\ldots,J, \tag{9}$$

$$\text{(ii)} \quad \max_{1 \leqslant j \leqslant J} \{\mathbb{E}[(\alpha_1^{(J)} \varepsilon_i^{(j)} + \beta_1^{(J)})^k]\} < 1. \tag{10}$$

The theorem states that the coefficients of regime $J, \alpha_1^{(J)}$ and $\beta_1^{(J)}$, are the key factor in determining the existence of moments for the TACD(1,1) process. The geometric ergodicity condition of Theorem 1 is sufficient for the existence of the first moment of the stationary distribution.

3. Modeling IBM transaction durations

This section provides a detailed discussion of our modeling approach for the IBM transaction duration data.[3] We first give a brief description of the IBM transaction data and then discuss specific issues including deterministic intraday patterns of the transaction durations and multiple transactions. Lastly, we consider the issues of model specification and estimation.

3.1. Data

We use the Trades, Orders Reports and Quotes (TORQ) data set that was put together by Joel Hasbrouck and the NYSE. The data set covers a 3 months period from November 1, 1990 to January 31, 1991 with 61 trading days. Each transaction contains a time stamp, measured in seconds after midnight, that reflects the time at which the transaction occurred. Also included are detailed information about each transaction including the volume, transaction price, and bid and ask quotes at the time of trade. Excluding the opening period of each day,[4] we obtain 51,363 transactions at 46,120 distinct transaction times for IBM.

Summary statistics for the IBM data are presented in Table 1. Similar to the finding of ER, we detect strong serial correlations in the durations. For example, the Ljung–Box statistic shows $Q(15)=9,520.89$ which uses 15 lags of autocorrelations and is highly significant. ER assume that such serial dependence can be modeled linearly. Here we allow for the possibility of nonlinear dependence. Since the standard deviation exceeds the mean, the data exhibit excess dispersion in the unconditional distribution. ER find that even after accounting for the temporal dependence neither the Exponential nor the Weibull distribution is capable of accounting this excess dispersion. Here we propose using the generalized gamma distribution. Lunde (1999) and Grammig and Maurer (2000) also extend the ACD model using generalized Gamma distribution and other distributions with non-monotonic hazard functions. Finally, approximately 10.21% of the IBM durations are zero seconds, indicating multiple transactions occurring within a given second. ER focus only on distinct transaction arrivals and, therefore, do not address the issue of zero duration. Multiple transactions within a second is indicative of rapid transaction and may therefore be useful in predicting the time until the next transaction. In this paper, we make use of such information.

[3] This allows close comparison with the results of ER. Modeling results for other stocks are available upon request.

[4] Since the open is sometimes delayed, and the opening auction results in extreme short durations around the open, a model of the trading process would be contaminated by including the opening trades. Thus, the opening period data, from 9:30 to 10:00 AM, have been removed.

Table 1
Summary statistics of IBM transaction durations[a]

	Mean	SD	Skewness	Excess kurtosis	Ljung–Box[b]
Non-zero observations (#*obs*. = 46120)					
Original duration	28.43	38.50	3.57	20.97	9520.89
Deseasonalized	1.032	1.374	3.501	20.998	7665.82
	Single zero	2-zero sequence	3-zero sequence	4-zero sequence	5-zero or more
Zero observations[c] (#*obs*. = 5243)					
Zeros	3995	443	52	26	18

[a]Transaction duration data in seconds from trades, orders reports and quotes (TORQ) data set. Data covers 11/01/1990–01/31/1991, and the data for 11/23/90 and 12/27/1990 have been removed because of periods of market closing. Opening period (9:30–10:00 AM) of each day has also been deleted.

[b]Ljung–Box statistics is given by $N(N+2)\sum_{j=1}^{L} r_j^2/(N-j)$, where N is the length of the series and r_j is the jth lag sample ACF. Degree of freedom L is 15 in the table. The statistic is distributed as χ^2_{15} with critical values: 25 (5%) and 30.58 (1%).

[c]Zero represents simultaneous transaction in the data. The table presents the distribution of consecutive zeros in the original transaction series. M-consecutive-zero sequence in the data means $M+1$ multiple transactions occured at one time point.

3.2. Daily seasonality

ER find that the frequency of transactions is higher near the open and close of the market. We adopt a two-step approach in accommodating this deterministic pattern in our model. First, using the duration data where $x_i^* = t_i - t_{i-1}$, we obtain a smoothed estimate of the daily deterministic component $\phi(t_{i-1}) \equiv \mathbb{E}[X_i^* | t_{i-1}]$ using local regression. This is simply the expected duration given the time of day the duration began. The daily seasonally adjusted data $\{x_i\}$ are obtained through $x_i = x_i^*/\phi(t_{i-1})$. Denote the deseasonalized duration series by X_i, where $\mathbb{E}[X_i] = 1$. In the second step, we apply the threshold ACD model to the adjusted series $\{x_i\}_{i=1}^N$.

We are effectively assuming that the daily deterministic component $\phi(\cdot)$ can be formulated as a multiplicative function such that $\mathbb{E}[X_i^* | X_{i-1}^*, \ldots, X_1^*] = \phi(t_{i-1})\psi_i(X_{i-1}, \ldots, X_1)$, where $\psi_i(X_{i-1}, \ldots, X_1) = \mathbb{E}[X_i | X_{i-1}, \ldots, X_1]$. ER estimate the deterministic pattern ($\phi(\cdot)$) and the ACD model ($\psi(\cdot)$) jointly. Since the information matrix is block diagonal between the parameters in $\phi(\cdot)$ and $\psi(\cdot)$, the proposed two-step approach provides consistent estimates as shown by Engle and Russell (1995). The difference between joint and two-step estimation becomes small when the sample size is large. Indeed, comparing the results in Engle and Russell (1995) with those in ER, the difference is negligible for the IBM data.

We use SPLUS function *supsmu* (Friedman, 1984) to compute daily seasonal component $\phi(t_{i-1})$. Consistent with previous studies we find the deterministic pattern is an inverted U shape with intensive trading around the opening and closing periods. Table 1 presents some summary statistics for the seasonally adjusted data. Comparing with the summary statistics for the original data in the table, the strong autocorrelations and excess dispersions in durations continue to hold even after adjusting for the time of day effect.

3.3. Multiple transactions

Let y_i be the number of transactions occurred at time t_i. Thus, $y_i > 1$ indicates that multiple transactions occurred within 1 s at time t_i. Occurrence of such transactions may be indicative of rapid transactions and may therefore impact the transaction intensity. We incorporate the information of multiple transactions into the TACD model. Since y_{i-1} is predetermined at the start of the ith duration, we can include it in the conditional mean of the ith transaction. Initially, we allow for two possibilities of the impact of multiple transactions on future transaction rates. First, the exact number of multiple transactions may carry information about future transaction rates. Second, the occurrence of multiple transactions, regardless of the number of transactions, may be informative about future transaction rates. To discriminate between these two hypotheses, we include two lag-1 indicator variables, $1_{\{Y_{i-1}>1\}}$ and $1_{\{Y_{i-1}>2\}}$ in the conditional mean equation of X_i. Our empirical results show that the coefficient of $1_{\{Y_{i-1}>2\}}$ is not significant, and hence the data support the second simple hypothesis. Consequently, it is the occurrence of multiple transactions that matters, not the exact number of multiple transactions. In what follows, we use $1_{\{Y_{i-1}>1\}}$ as a regressor in the specification of the conditional mean duration ψ_i.

3.4. Model specification, estimation and diagnostics

For our application to financial transactions data we begin with a 3-regime TACD(1,1) model with generalized gamma distribution as the error distribution and an indicator of multiple transactions (zeros) as a regressor in the mean equation. The generalized gamma distribution allows for wider flexibility in the hazard function of the trading process.[5] A three-parameter

[5] See Lancaster (1990) for a more complete description.

generalized gamma density is given by

$$f(x|\gamma,\kappa,\lambda)=\frac{\gamma x^{\kappa\gamma-1}}{\lambda^{\kappa\gamma}\Gamma(\kappa)}\exp\left\{-\left(\frac{x}{\lambda}\right)^{\gamma}\right\},\quad x>0. \tag{11}$$

The dynamics for the conditional mean is specified as

$$\psi_i=\begin{cases}\alpha_0^{(1)}+\alpha_1^{(1)}X_{i-1}+\beta_1^{(1)}\psi_{i-1}+\omega^{(1)}\mathbf{1}_{\{Y_{i-1}>1\}} & \text{if } 0<X_{i-1}\leqslant r_1,\\ \alpha_0^{(2)}+\alpha_1^{(2)}X_{i-1}+\beta_1^{(2)}\psi_{i-1}+\omega^{(2)}\mathbf{1}_{\{Y_{i-1}>1\}} & \text{if } r_1<X_{i-1}\leqslant r_2,\\ \alpha_0^{(3)}+\alpha_1^{(3)}X_{i-1}+\beta_1^{(3)}\psi_{i-1}+\omega^{(3)}\mathbf{1}_{\{Y_{i-1}>1\}} & \text{if } r_2<X_{i-1}<\infty.\end{cases} \tag{12}$$

We also allow the shape parameter κ of the generalized gamma distribution to be different across regimes. Let $\theta^*=(\gamma,\Gamma_1,\Gamma_2,\Gamma_3)$, where $\Gamma_j=(\alpha_0^{(j)},\alpha_1^{(j)},\beta_1^{(j)},\omega^{(j)},\kappa^{(j)})$ for $j=1,2,3$, and $\theta=(\theta^*,r_1,r_2)$. The parameter space Θ of our model can be written as $\Theta=\{\theta:\gamma>0;\ \alpha_0^{(j)}>0,\ \alpha_1^{(j)}>0,\ \beta_1^{(j)}>0,\ \kappa^{(j)}>0,\ j=1,2,3;\ 0<r_1<r_2\}$.

The log likelihood function for the ith observation is expressed as

$$\ell_i=\sum_{j=1}^{3}\mathbf{1}_{\{x_{i-1}\in R_j\}}\left[\log(\gamma)+(\kappa^{(j)}\gamma-1)\log\left(\frac{x_i}{\lambda_i^{(j)}}\right)\right.$$
$$\left.-\log(\lambda_i^{(j)}\Gamma(\kappa^{(j)}))-\left(\frac{x_i}{\lambda_i^{(j)}}\right)^{\gamma}\right], \tag{13}$$

where $\lambda_i^{(j)}=\psi_i\Gamma(\kappa^{(j)})/\Gamma(\kappa^{(j)}+1/\gamma)$, if $x_{i-1}\in R_j$, and ψ_i is given by Eq. (12). Maximum likelihood estimates can then be obtained by maximizing the conditional log-likelihood function $\log\mathscr{L}(\theta^*|x_1,\ldots,x_N,y_1,\ldots,y_N;r_1,r_2)=\sum_{i=2}^{N}\ell_i$. Using a grid search across r_1 and r_2, we numerically maximize the likelihood for each pair using the BHHH algorithm (Berndt et al., 1974). The MLE is the pair of r_1 and r_2 and the corresponding θ^* that produces the highest maximized value of the likelihood.

It is unlikely that the dynamic relationship between the last observation of a day and the subsequent open the following day is the same as the within day relationship. Following ER, we re-initialize the process each day by setting the conditional expectation of the first waiting time after 10:00 AM to the average (daily seasonally adjusted) duration over the 10 min prior to 10:00 AM.

Estimation results for the full sample are presented in Panel A of Table 2. The parameters of each regime are significant at the 5% level. Since the shape parameter κ is significantly different across regimes, the hazard functions are distinct for each regime. The sums of α_1 and β_1 are 1.077, 1.005 and 0.963, respectively, for regimes 1, 2 and 3, indicating high persistence for each regime but the process is still ergodic since $\alpha_1^{(3)}+\beta_1^{(3)}<1$.

Panel B of Table 2 shows diagnostic checks of the 3-regime TACD(1,1) model using a nonlinear test, a variance test, and the Ljung–Box test based

Table 2
Model results for IBM: full sample

Panel A: parameter estimates of TACD(1,1) model[a]

	$r_1 = 0.188$ (25%); $r_2 = 0.947$ (65%)					
	Regime 1		Regime 2		Regime 3	
Parameter	Estimate	t-stat	Estimate	t-stat	Estimate	t-stat
α_0	0.015	2.692	0.024	3.203	0.088	10.821
α_1	0.206	3.654	0.081	5.926	0.036	11.356
β_1	0.871	145.829	0.924	151.166	0.927	126.485
ω	−0.026	−5.772	−0.066	−7.157	−0.075	−5.976
γ	0.364	24.452	0.364	24.452	0.364	24.452
κ	5.254	12.391	5.880	12.590	5.117	12.562

Panel B: tests of residuals

Nonlinear test		Variance test		Ljung–Box statistics			
				1st Moment		2nd Moment	
Statistic	P-value	Statistic	P-value	Statistic	P-value	Statistic	P-value
2.844	0.000006	−5.558	0.000000	30.308	0.011	18.181	0.253

[a]The 3-regime TACD(1,1) model is given by

$$\begin{cases} X_i/\psi_i \equiv \varepsilon_i^{(j)} \sim \mathscr{G}(\gamma, \kappa^{(j)}) \\ \psi_i = \alpha_0^{(j)} + \alpha_1^{(j)} X_{i-1} + \beta_1^{(j)} \psi_{i-1} + \omega^{(j)} \mathbf{1}_{\{Y_{i-1} > 1\}} \end{cases} \quad \text{if } X_{i-1} \in R_j$$

for $j = 1, 2, 3$, where $\psi_i \equiv \mathbb{E}[X_i | X_{i-1}, \ldots, X_1; \theta_1]$ and $\mathscr{G}(\gamma, \kappa)$ is a standardized generalized gamma distribution with mean equal to one. With $\lambda = \Gamma(\kappa)/\Gamma(\kappa + 1/\gamma)$, the probability density is given by

$$f(x|\gamma, \kappa) = \frac{\gamma x^{\kappa\gamma - 1}}{\lambda^{\kappa\gamma} \Gamma(\kappa)} \exp\left\{-\left(\frac{x}{\lambda}\right)^{\gamma}\right\}, \quad x > 0.$$

on the residuals of the model.[6] The nonlinear test is used to examine the nonlinear dependence between the residuals and the past information set. The reported test statistic of 2.844 is significant at 0.00001 level suggesting remaining nonlinearities in the residuals. The variance test is designed to test for the null hypothesis of no excess dispersion required under correct specification. The test statistic for the model is −5.558 with a P-value close to zero. The negative sign of the test statistic shows the residuals of the model are under dispersed. The Ljung–Box statistic for the first 15 lags of the residuals is 30.308, which is marginal at the 1% level. We also examine the Ljung–Box statistic for the second moment of the residuals which is well

[6] For details of the nonlinear test and the variance test, please refer to page 1143–4 of ER.

below the 1% critical value with a P-value of 0.253. This indicates that the 3-regime TACD(1,1) can explain most of the temporal dependence in the transaction durations. [7]

The nonlinear test and the variance test suggest that there are still significant nonlinearities and under dispersions that the proposed TACD(1,1) model cannot fully accommodate. As we will examine in the next section, structural breaks play a key factor. After detecting important structural breaks in the sampling period, the proposed model can pass all three of the above tests.

4. Structural breaks

Stationarity is often a central assumption when making inference in time series models. This is also true for our analysis of the stock transaction process. Throughout the trading process, there are information events like periodic news announcement or unexpected news events which may have a significant influence on the trading process. It is natural to ask: how long will these shocks persist? Is there any structural change due to big news events? In this section, we examine how plausible the stationarity assumption is for the IBM data.

4.1. Structural breaks of IBM transaction durations

We adopt a sequential parametric approach to detect potential structural breaks in the data. The model we use is the 3-regime TACD(1,1) model which as discussed earlier is quite flexible in accommodating nonlinearities and excess dispersions in the data. The Lagrange Multiplier type tests of Andrews and Ploberger (1994) and Andrews (1993), namely SupLM, ExpLM and AveLM, are performed to detect unknown change points in the parametric model. We restrict our attention to break points between consecutive trading days, that is, we treat the data from the same day as a block unit. Our method is sequential in the sense that once we find a break point, we will continue to conduct the tests for each subsample until there is no break point in each partitioned subsample. [8]

[7] We also estimate a 3-regime TACD(2,2) model which can well pass the Ljung–Box statistic at 5% level but still fails the nonlinear test and the variance test. This confirms that we need to search for the reason, other than the specification of the dynamics, which result in failing the nonlinear test and the variance test.

[8] The structural breaks discussed here signify the possibility of nonstationarity and changes in trading behavior in an intraday micro-scale. As pointed out by an anonymous referee, the power and significance levels of the sequential test procedure need to be studied. This can be done using Monte Carlo methods. Here we use the sequential test as a guide and indication for nonstationary behavior. We leave a more comprehensive study for future research.

We find six break points. At the first stage, we run the test for the whole sample, the three test statistics are 502.43, 247.121 and 163.054, respectively, for SupLM, ExpLM and AveLM. The corresponding critical values at the 1% significant level are 44.20, 17.50 and 24.66. This indicates a significant structural change starting on January 17, 1991, the date with the highest LM test statistic. We then run the test for the two subsamples separated by January 17, 1991 and detect two more break points on December 10, 1990 and January 22, 1991. Thus, after the second stage of our procedure, three break points have been detected, and the sample is divided into four subsample periods. Repeating this procedure until the test statistics are no longer significant, we eventually identify six break points in the series and divide the sample into seven subperiods, for which we will label them as period 1 to 7, chronologically. Fig. 1 shows this binary branching process in a diagram: each node corresponds to a subperiod of the sample and summarizes the tests of SupLM, ExpLM and AveLM with 1% critical values given in brackets. The seven end nodes of the binary tree correspond to the final seven subperiods obtained.

The detected structural changes can be closely aligned with real economic events.[9] In order to show that the structural breaks are not only economically meaningful but also statistical meaningful, we shall explore the structural differences across subperiods implied by the estimated 3-regime TACD(1,1) models.

4.2. Exploring structural differences

The seven subperiods are quite different statistically. Table 3 gives some summary statistics of transaction durations for each subperiod. We see clearly that the three short subperiods, Periods 2, 4, and 6, are obviously transacting at a faster pace than the other four subperiods. There are not only shifts in the mean, but also shifts in the variance. The three short subperiods also have lower variances. Temporal dependence for each subperiod is still strong, as shown by the Ljung–Box statistics.

Model estimation results for each subsample are presented in Table 4. They provide an opportunity to examine closely the properties of each subperiod. The three short and news event dominated subperiods have distinct dynamics and shapes as compared to that of the other four subperiods. Significant differences among the unconditional moments and autocorrelation functions implied by the fitted models of each subperiod provide further concrete and intuitive evidence about the detected structural changes. Since analytical

[9] The three short subperiods, Periods 2, 4 and 6, are corresponding to IBM's announcement of division executive changes, rumors about IBM's earnings outlook, and IBM's earnings announcement, respectively.

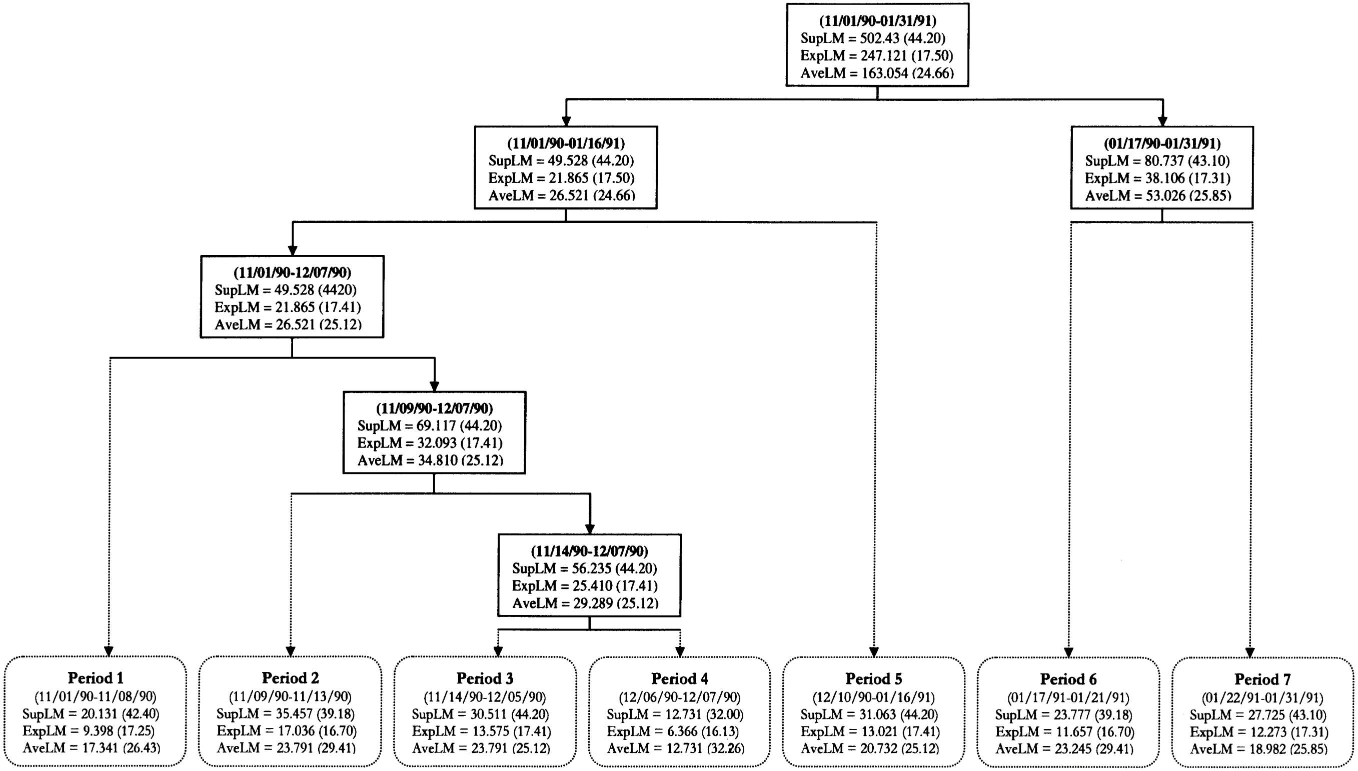

Fig. 1. Diagram of structure break test procedures for IBM transaction duration. We use the test statistics to sequentially partition the sample period until that the test statistic (SupLM) is not significant in each sub-period. The parametric model used to conduct test is the 3-regime GACD(1, 1) model. Critical values in brackets are at %1 level. Values for SupLM are from Andrews (1993) and values for ExpLM and AveLM are from Andrews and Ploberger (1994).

Table 3
Sub-period summary statistics of IBM transaction durations[a]

Period	NumObs	Min	1st quantile	Median	3rd quantile	Max
1: (11/01/90–11/08/90)	3774	0.0315	0.221	0.720	1.659	15.980
2: (11/09/90–11/13/90)	3121	0.0315	0.143	0.423	0.979	9.746
3: (11/14/90–12/05/90)	9449	0.0315	0.221	0.676	1.540	15.820
4: (12/06/90–12/07/90)	2025	0.0315	0.158	0.420	1.071	7.838
5: (12/10/90–01/16/91)	14645	0.0315	0.250	0.757	1.728	25.170
6: (01/17/91–01/21/91)	4400	0.0315	0.126	0.314	0.725	7.234
7: (01/22/91–01/31/91)	8706	0.0315	0.159	0.441	0.966	16.740

Period	NumObs	Mean	SD	Skewness kurtosis	Excess	Ljung–Box[b] (#lag = 15)
1: (11/01/90–11/08/90)	3774	1.247	1.557	2.940	13.688	103.004
2: (11/09/90–11/13/90)	3121	0.754	0.963	3.070	14.600	399.625
3: (11/14/90–12/05/90)	9449	1.163	1.425	2.808	12.318	451.845
4: (12/06/90–12/07/90)	2025	0.774	0.951	2.606	9.431	123.891
5: (12/10/90–01/16/91)	14645	1.322	1.690	3.218	17.248	1109.078
6: (01/17/91–01/21/91)	4400	0.535	0.627	2.723	12.195	510.743
7: (01/22/91–01/31/91)	8706	0.721	0.868	3.451	26.754	574.171

[a]Transaction duration data in seconds from trades, orders reports and quotes (TORQ) data set. Data covers 11/01/1990–01/31/1991, and the data for 11/23/90 and 12/27/1990 have been removed because of periods of market closing. Opening period (9:30–10:00 AM) of each day has also been deleted.

[b]Ljung–Box statistics is given by $N(N+2)\sum_{j=1}^{L} r_j^2/(N-j)$, where N is the length of the series and r_j is the jth lag sample ACF. Degree of freedom L is 15 in the table. The statistic is distributed as χ^2_{15} with critical values: 25 (5%) and 30.58 (1%).

characterizations of the stationary distribution and autocorrelation structure are not available for the threshold ACD model, we adopt the parametric bootstraps approach of Tsay (1992) to obtain the unconditional moments and autocorrelations numerically.

For each estimated model, we simulate 100 series with a length of 50,000 observations, and then extract the sample moments and autocorrelations for each of the 100 series.[10] Table 5 shows the simulated moments from the estimated models for each subperiod. The first two moments are significantly different across the seven subperiods. Thus, our models for the seven subperiods are distinct in terms of the unconditional moments.

[10] Since we do not have a structural model about the arrival of multiple transactions, we simulate the $\{Y_i^0\}$ sequence as i.i.d. Bernoulli random variable with the sample proportion of multiple transactions as the probability of success. We also simulate the $\{Y_i^0\}$ sequence with other schemes, and the results are quite similar.

Table 4
Parameter estimates of models for IBM[a]

Sub-period	Parameter	Regime 1 Estimate	Regime 1 t-stat	Regime 2 Estimate	Regime 2 t-stat	Regime 3 Estimate	Regime 3 t-stat
Period 1	α_0	0.027	0.684	0.129	3.951	0.783	2.070
(11/01/90–11/08/90)	α_1	0.205	0.402	0.054	3.440	0.021	0.636
$r_1 = 0.172(20\%)$	β_1	0.851	30.073	0.887	32.970	0.405	1.527
$r_2 = 4.201(95\%)$	ω	−0.102	−2.431	−0.218	−5.162	−0.121	−0.568
	γ	0.414	7.849	0.414	7.849	0.414	7.849
	κ	3.813	4.003	4.004	4.103	3.260	3.603
Period 2	α_0	0.024	1.513	0.056	1.967	0.225	3.777
(11/09/90–11/13/90)	α_1	0.280	1.419	0.039	0.786	0.034	1.868
$r_1 = 0.175(30\%)$	β_1	0.802	33.468	0.928	32.235	0.764	12.019
$r_2 = 0.979(75\%)$	ω	−0.048	−2.552	−0.085	−2.336	−0.039	−0.640
	γ	0.266	4.452	0.266	4.452	0.266	4.452
	κ	10.171	2.239	11.203	2.257	9.518	2.245
Period 3	α_0	0.044	2.099	0.002	0.087	0.378	6.688
(11/14/90–12/05/90)	α_1	−0.081	−0.326	0.103	3.772	0.029	3.136
$r_1 = 0.171(20\%)$	β_1	0.857	48.578	0.948	58.909	0.710	15.862
$r_2 = 1.307(70\%)$	ω	−0.038	−2.079	−0.099	−3.544	−0.105	−2.203
	γ	0.423	12.378	0.423	12.378	0.423	12.378
	κ	3.683	6.257	4.183	6.479	3.839	6.416
Period 4	α_0	0.015	1.711	−0.194	−0.788	0.187	2.567
(12/06/90–12/07/90)	α_1	0.035	1.662	0.083	0.456	−0.008	−0.526
$r_1 = 1.071(75\%)$	β_1	0.957	78.018	1.188	13.413	0.811	11.650
$r_2 = 1.462(85\%)$	ω	−0.028	−1.652	−0.178	−2.882	0.161	2.222
	γ	0.291	3.593	0.291	3.593	0.291	3.593
	κ	8.847	1.835	7.364	1.785	7.891	1.803
Period 5	α_0	0.026	1.537	0.073	2.625	0.176	7.381
(12/10/90–01/16/91)	α_1	0.110	0.951	0.031	0.851	0.030	5.109
$r_1 = 0.250(25\%)$	β_1	0.883	71.133	0.908	56.840	0.892	50.646
$r_2 = 1.045(60\%)$	ω	−0.063	−2.763	−0.105	−3.646	−0.072	−2.261
	γ	0.438	16.665	0.438	16.665	0.438	16.665
	κ	3.564	8.590	3.923	8.648	3.458	8.720
Period 6	α_0	0.044	2.046	0.040	4.152	0.047	0.788
(01/17/91–01/21/91)	α_1	−0.141	−0.505	0.072	4.440	0.021	1.067
$r_1 = 0.096(20\%)$	β_1	0.819	29.537	0.894	49.277	0.953	13.288
$r_2 = 1.301(90\%)$	ω	−0.026	−2.177	−0.055	−4.315	−0.078	−1.920
	γ	0.222	4.076	0.222	4.076	0.222	4.076
	κ	17.947	2.033	17.551	2.060	14.687	2.024
Period 7	α_0	0.025	2.668	0.060	4.102	0.204	3.544
(01/22/91–01/31/91)	α_1	0.153	1.405	0.068	3.872	0.024	1.809
$r_1 = 0.190(30\%)$	β_1	0.864	58.591	0.889	47.635	0.778	11.175
$r_2 = 1.358(85\%)$	ω	−0.042	−5.156	−0.093	−5.552	−0.076	−2.144
	γ	0.286	8.132	0.286	8.132	0.286	8.132
	κ	9.081	4.063	10.335	4.142	9.099	4.095

[a]The 3-regime TACD(1,1) model is given by

$$\begin{cases} X_i/\psi_i \equiv \varepsilon_i^{(j)} \sim \mathscr{G}(\gamma, \kappa^{(j)}) \\ \psi_i = \alpha_0^{(j)} + \alpha_1^{(j)} X_{i-1} + \beta_1^{(j)} \psi_{i-1} + \omega^{(j)} \mathbf{1}_{\{Y_{i-1}>1\}} \end{cases} \quad \text{if } X_{i-1} \in R_j$$

for $j = 1, 2, 3$, where $\psi_i \equiv \mathbb{E}[X_i | X_{i-1}, \ldots, X_1; \theta_1]$ and $\mathscr{G}(\gamma, \kappa)$ is a standardized generalized gamma distribution with mean equal to one. With $\lambda = \Gamma(\kappa)/\Gamma(\kappa + 1/\gamma)$, the probability density is given by

$$f(x|\gamma,\kappa) = \frac{\gamma x^{\kappa\gamma-1}}{\lambda^{\kappa\gamma}\Gamma(\kappa)} \exp\left\{-\left(\frac{x}{\lambda}\right)^{\gamma}\right\}, \quad x > 0.$$

Table 5
Simulated moments using models estimated for each sub-period

	1st moment	SE	2nd moment	SE	3rd moment	SE	4th moment	SE
Period 1	1.292	0.001125	4.854	0.0114	38.706	0.304	551.086	15.332
Period 2	0.833	0.000886	2.125	0.0070	13.053	0.177	164.491	7.705
Period 3	1.219	0.001351	4.132	0.0106	28.876	0.200	353.210	7.869
Period 4	0.788	0.000836	1.836	0.0058	10.214	0.196	127.752	11.853
Period 5	1.356	0.001845	5.280	0.0178	43.907	0.520	682.345	43.052
Period 6	0.553	0.000628	0.852	0.0027	3.127	0.039	23.946	0.939
Period 7	0.741	0.000721	1.507	0.0041	6.847	0.089	63.499	4.233

Fig. 2 shows the 90% confidence bands for autocorrelations of the subperiods. Autocorrelations are quite different between each pair of neighboring subperiods. For example, Period 2 obviously has a higher level of autocorrelations compared with that of Period 1 and thus has a more intensive clustering of transaction arrivals. Those autocorrelations clearly show that the dynamics of the estimated models for the seven subperiods are distinct.

4.3. Model diagnostics

Structural breaks can be one of the sources of rejection of the linear ACD model in favor of the nonlinear model. We now examine whether the nonlinear specification remains necessary after breaking the sample into subperiods.

We use the nonlinear test, the variance test and the Ljung–Box test of residuals to compare the goodness of fit of the proposed 3-regime TACD(1,1) model and the best model in ER, the linear ACD(2,2) model with Weibull errors (WACD(2,2)), for each subperiod. Table 6 shows the test results. The linear WACD(2,2) model has difficulties to capture the nonlinearities in the transaction durations, especially for Periods 3, 5 and 7, for which the P-values are smaller than 0.005. The 3-regime TACD(1,1) model absorbs most of the nonlinearities in the transaction durations. The P-values of the nonlinear test are all bigger than 0.05 for each subperiod, indicating good fit in terms of modeling nonlinearity. The results of the variance test show a clear pattern of over dispersion for residuals of the linear WACD(2,2) model and under dispersion for residuals of the 3-regime TACD(1,1) model. While residuals of the nonlinear model passed the variance test at the 1% level for five out of seven subperiods, the residuals of the linear model can only pass in two subperiods and show severe over dispersion for Periods 5, 6 and 7. The residuals of both the 3-regime TACD(1,1) and the linear WACD(2,2) model passed the Ljung–Box test of autocorrelations for the first two moments. It indicates that both models can capture the linear intertemporal dependence in

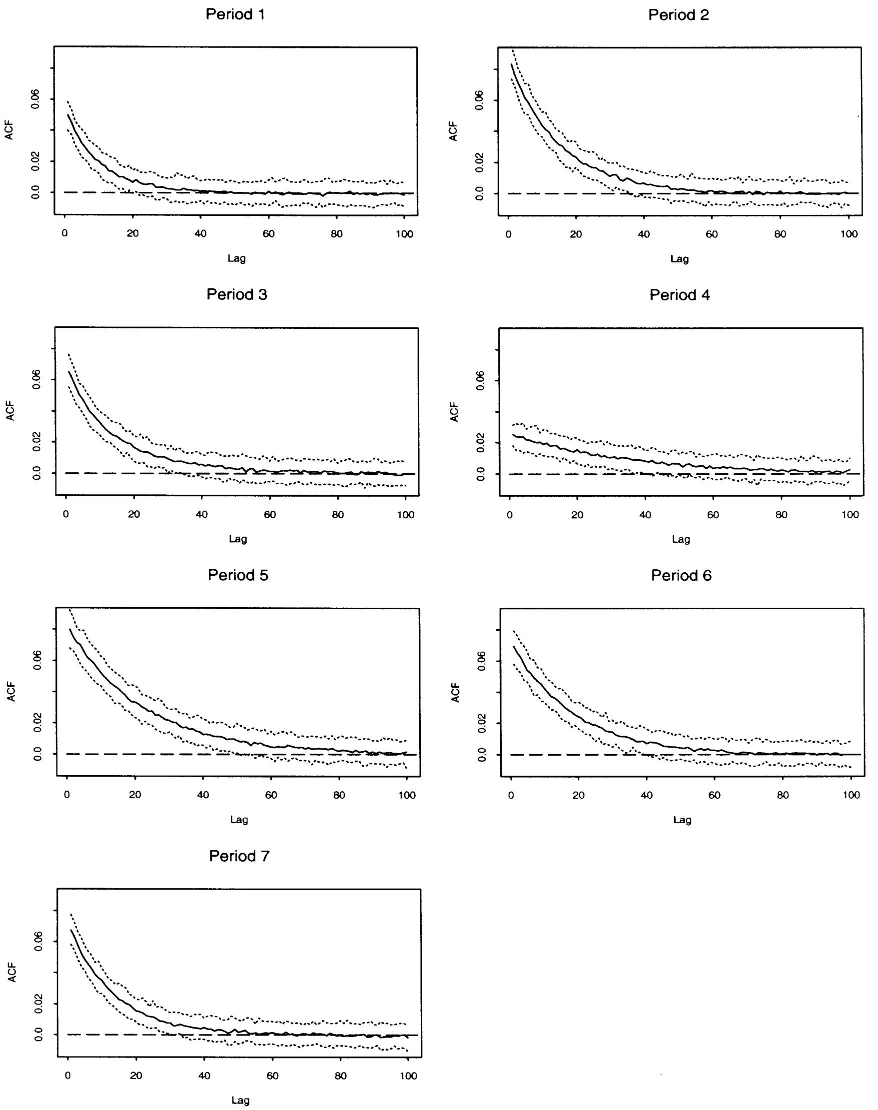

Fig. 2. Autocorrelations from simulated data using models estimated for each sub-period. Areas between two dash curves are 90% confidence band for the autocorrelations. The solid curve is the median level of the autocorrelations.

the transaction durations, although nonlinear dependence remains in the linear WACD(2,2). In summary, we find that even after we identify the six break points we still reject strongly the simple linear ACD model in favor of the nonlinear TACD(1,1) specification.

Table 6
Tests of residuals[a] for IBM data

	Nonlinear test		Variance test		Ljung–Box statistics			
					1st moment		2nd moment	
	Statistic	P-value	Statistic	P-value	Statistic	P-value	Statistic	P-value
Period 1 (11/01/90–11/08/90)								
3-regime TACD(1, 1)	0.878	0.630	−1.728	0.0839	8.384	0.907	12.537	0.637
WACD(2, 2)	1.345	0.125	2.519	0.0118	6.829	0.962	10.387	0.795
Period 2 (11/09/90–11/13/90)								
3-regime TACD(1, 1)	1.082	0.358	−1.664	0.096	13.653	0.552	30.469	0.0103
WACD(2, 2)	1.802	0.0109	3.408	0.0007	11.876	0.688	14.978	0.453
Period 3 (11/14/90–12/05/90)								
3-regime TACD(1, 1)	1.403	0.0952	−2.662	0.0078	15.338	0.427	18.342	0.245
WACD(2, 2)	2.698	0.0000	3.815	0.0001	11.941	0.684	13.055	0.598
Period 4 (12/06/90–12/07/90)								
3-regime TACD(1, 1)	1.014	0.443	−1.457	0.145	10.486	0.788	17.905	0.268
WACD(2, 2)	1.192	0.240	2.210	0.0271	8.214	0.915	4.503	0.996
Period 5 (12/10/90–01/16/91)								
3-regime TACD(1, 1)	1.142	0.288	−2.593	0.0095	17.313	0.300	13.528	0.562
WACD(2, 2)	1.941	0.0044	5.783	0.0000	25.159	0.0479	9.953	0.823
Period 6 (01/17/91–01/21/91)								
3-regime TACD(1,1)	0.810	0.723	−2.06	0.0394	23.403	0.0759	12.627	0.631
WACD(2, 2)	1.441	0.0792	4.502	0.0000	14.695	0.474	8.527	0.901
Period 7 (01/22/91–01/31/91)								
3-regime TACD(1, 1)	0.992	0.471	−2.575	0.0100	11.082	0.746	30.721	0.0096
WACD(2, 2)	2.182	0.0009	6.508	0.0000	9.687	0.839	12.128	0.669

[a]Residuals are defined by integrated hazards. The *integrated hazard* of X_i is defined as $I_i(X_i) = \int_0^{X_i} h_i(s)\,ds$, where $h_i(\cdot)$ is the hazard function of X_i.

5. Economic implications of the threshold dynamics

We explore next the economic meaning of the regimes and see whether our model can shed some light on better understanding of the trading process. Our approach is to relate the behavior of economic variables across regimes to the predictions of market microstructure theories. We first develop the theoretical basis for our analysis from the market microstructure literature, and then show the behavior of those economic variables across regimes.

5.1. Market microstructure theories

Information-based market microstructure theory focuses on how the market evolves following the arrival of new information, especially, how new information is absorbed into market prices. If there are asymmetrically informed traders, the specialist must infer private information from the action of the traders. Hence, the trading process as a source of information to asymmetrically informed market participants plays an important role in price formation. Information-based microstructure theory has four important predictions, about the extent of information asymmetry in the trading process and the behavior of transaction rates, bid-ask spreads, volume and volatility, respectively. First, Easley and O'Hara (1992b) develop the idea of trade arrival as a signal to market participants and show that higher transaction rates are associated with a higher likelihood of informed trading. Second, the theoretical results of Glosten and Milgrom (1985) and Easley and O'Hara (1992b) imply that specialists set the bid and ask quotes to reflect the possibility of trading with an informed agent and will widen the spreads when the extent of information asymmetry is increasing. Hence, we expect that the higher the likelihood of informed trades the wider the spreads are. Third, informed traders have the incentive to trade a large quantity of the stock at a fixed price given their superior information, as suggested by Easley and O'Hara (1987), Easley and O'Hara (1992a) and Lee et al. (1993). Thus, the likelihood of informed trading is positively correlated with trading volume. Lastly, the price volatility is also positively correlated with the extent of information asymmetry. Market makers will update their bid and ask quotes more rapidly once they learn through the sell and buy orders that the proportion of informed traders is high. Based on these predictions, we shall examine whether transaction rates move together with spread, volume and volatility across regimes in our model.

5.2. *Exploring economic meanings of regimes*

We summarize the behavior of bid-ask spreads, transaction volume per second and price volatility across the three regimes for each of the seven subperiods in Table 7. The transactions in the fast regime of our model have the largest spreads, volume (except for Periods 1, 4 and 6) and volatility, and those in the third and slow regime have the lowest. Thus, high transaction rates, wider spreads, larger volume and higher price variation occur around the same time suggesting fast transacting regime may be indicative of informed trading. Our findings of transaction rates move together with spread, volume and volatility are consistent with the market microstructure theories mentioned above. This further suggests that we can characterize the fast transacting regime as the informed trading regime and the slow transacting regime as non-informed trading regime. The middle regime is the 'normal' trading regime, which represents an equilibrium state with mixture of informed trading and noninformed trading and, as expected, most of the observations fall into this category.

Table 7
Summary of economic variables across regimes for IBM data[a]

			Spread[b]		Volume/Sec[c]		Volatility[d]	
		Num. Obs.	Mean	SE of mean	Mean	SE of mean	Mean	SE of mean
Period 1	Regime 1	756	0.183	0.00248	4.728	0.525	0.0566	0.00696
(11/01/90–11/08/90)	Regime 2	2829	0.176	0.00121	4.295	0.309	0.0294	0.00172
	Regime 3	189	0.159	0.00407	4.986	1.137	0.0257	0.00642
Period 2	Regime 1	937	0.178	0.00221	7.390	0.773	0.0523	0.00664
(11/09/90–11/13/90)	Regime 2	1404	0.171	0.00167	5.065	0.591	0.0215	0.00240
	Regime 3	780	0.162	0.00213	5.444	0.561	0.0152	0.00260
Period 3	Regime 1	1891	0.178	0.00158	6.125	0.531	0.0612	0.00617
(11/14/90–12/05/90)	Regime 2	4723	0.174	0.00096	3.891	0.212	0.0287	0.00156
	Regime 3	2835	0.160	0.00109	4.106	0.301	0.0221	0.00198
Period 4	Regime 1	1519	0.194	0.00190	5.875	0.350	0.0541	0.00424
(12/06/90–12/07/90)	Regime 2	202	0.176	0.00468	8.330	1.892	0.0324	0.00892
	Regime 3	304	0.169	0.00363	5.406	0.808	0.0277	0.00551
Period 5	Regime 1	3662	0.189	0.00156	4.577	0.297	0.0541	0.00334
(12/10/90–01/16/91)	Regime 2	5125	0.178	0.00103	3.357	0.166	0.0281	0.00135
	Regime 3	5858	0.164	0.00078	4.146	0.504	0.0201	0.00203
Period 6	Regime 1	881	0.180	0.00242	5.678	0.598	0.0495	0.00598
(01/17/91–01/21/91)	Regime 2	3079	0.180	0.00125	5.661	0.391	0.0342	0.00222
	Regime 3	440	0.163	0.00280	5.821	0.795	0.0243	0.00454
Period 7	Regime 1	2613	0.184	0.00134	5.444	0.376	0.0479	0.00348
(01/22/91–01/31/91)	Regime 2	4788	0.173	0.00092	4.536	0.254	0.0235	0.00122
	Regime 3	1305	0.159	0.00155	5.176	0.502	0.0172	0.00203

[a]The parametric model we use is a 3-regime TACD(1,1) model, which is given by Eq. (12).
[b]Difference between market prevailing bid and ask prices.
[c]Daily deseasonalized transaction volume divided by daily deseasonalized transaction duration.
[d]Absolute value of changes in the midpoint of prevailing quotes divided by daily deseasonalized transaction duration.

To formally test our conclusion of regime identification, we run the following logistic regression for each subperiod:

$$\log\left(\frac{\mu_i}{1-\mu_i}\right) = b_0 + b_1\text{Spread}_i + b_2\text{Volume/Sec}_i + b_3\text{Volatility}_i, \qquad (14)$$

where $\mu_i = \mathbb{E}[Z_i]$ and Z_i is a Bernoulli variable indicating whether the ith observation belongs to regime 1, that is, $Z_i = \mathbf{1}_{\{X_{i-1}\in R_1\}}$. Table 8 presents parameter estimates of the logistic regressions. Price volatility is most significant among all the regressors, the coefficient of which is positive and significant for each subperiod. Spreads are also significantly and positively related to the

Table 8
Parameter estimates of logistic regressions[a]

	b_0 (Intercept)		b_1 (Spread)		b_2(Volume/Sec)		b_3 (Volatility)	
	Estimate	t-stat	Estimate	t-stat	Estimate	t-stat	Estimate	t-stat
Period 1	−1.705	−14.341	0.247	2.281	0.00153	0.654	1.535	4.706
Period 2	−1.263	−11.156	0.339	3.188	0.00446	2.413	1.747	5.073
Period 3	−1.770	−24.263	0.304	4.454	0.00616	4.410	1.418	7.405
Period 4	0.306	2.090	0.727	5.386	−0.00285	−0.912	1.256	2.592
Period 5	−1.612	−32.447	0.453	10.129	0.00073	1.080	1.404	8.473
Period 6	−1.464	−14.009	0.048	0.508	−0.00027	−0.142	0.762	3.057
Period 7	−1.392	−20.378	0.477	7.583	0.00215	1.726	1.611	7.346

[a]The logistic regression is given by

$$\log\left(\frac{\mu_i}{1-\mu_i}\right) = b_0 + b_1\,\text{Spread}_i + b_2\text{Volume/Sec}_i + b_3\,\text{Volatility}_i,$$

where $\mu_i = \mathbb{E}[Z_i]$ and $Z_i = 1_{\{X_{i-1} \in R_1\}}$.

likelihood of being in regime 1 for all subperiods but Period 6. The results for volume per second are mixed, as the coefficients are negative in Periods 4 and 6. Excluding Periods 4 and 6, our regression results strongly support the conclusion that higher probability in regime 1 is associated with wider spreads, higher volume and higher volatility, and, hence, higher likelihood of informed trading. We do not have a clear explanation for Periods 4 and 6, but we know that Period 4 is a short 2-day period dominated by rumors about IBM's earnings outlook and Period 6 is complicated by the IBM's earnings announcement and the start of Gulf war.

5.3. Understanding the nonlinear dynamics

We now conduct a simple analysis of the nonlinear dynamics of the TACD model used to examine the type of nonlinearity in the trading process.

The sum of α and β determines the stability of the process, and with different $\alpha+\beta$ across regimes, the model shows different characteristics of the trading process at each regime. Based on the fitted models for subperiods, $\alpha_1^{(1)} + \beta_1^{(1)} > 1$ for three subperiods. This is a standard example of threshold model that subsystem nonstationarity does not mean system nonstationarity. Theorem 1 shows that the process is stationary since $\alpha_1^{(3)} + \beta_1^{(3)} < 1$. Considering the first regime as an information regime, $\alpha_1^{(1)} + \beta_1^{(1)} > 1$ means once the process enters the fast transaction regime, there is a force to magnify the duration in an exponential speed. Viewing the fast transaction regimes as an informed regime this suggests that the effect of the information dies out in exponential rate. The third regime's $\alpha_1 + \beta_1$ is relative low, this ensures that

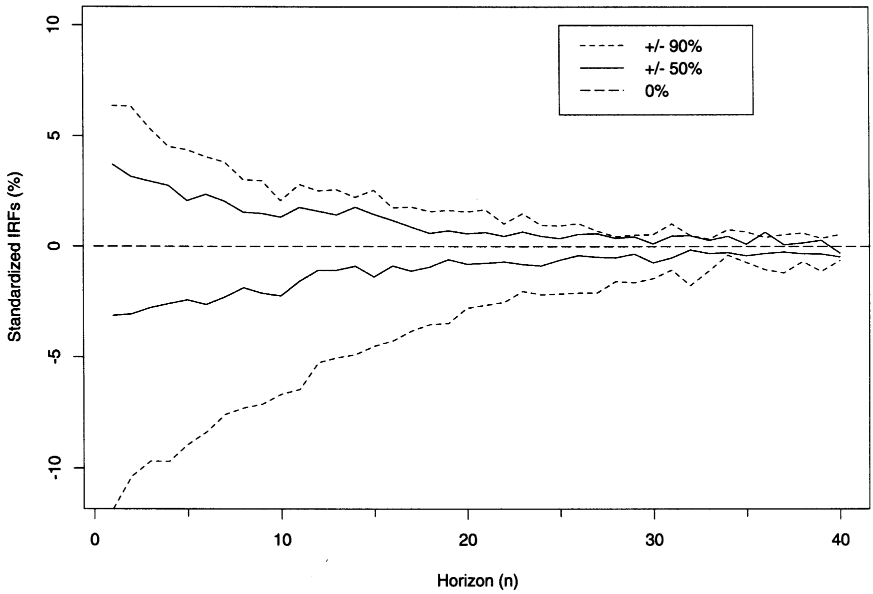

Fig. 3. Impulse response function for Period 7. The settings are: $\psi_i = 0.741$, $x_i = 0.741$, and $(r_1, r_2) = (0.190, 1.358)$. Two pairs of shocks are used: $\delta = \pm 50\%$, and $\delta = \pm 90\%$.

the process won't stay too long in the slow trading mode. Thus, the model indicates rapid exit from the two extreme regimes once the process enters them.

The propagation of shocks in the proposed TACD model is path dependent and asymmetric between positive and negative shocks. We use impulse response function (IRF) to illustrate the asymmetry of the proposed model. The standardized IRFs represent the percentage difference in conditional mean duration caused by the shock. Fig. 3 shows the standardized IRFs for the next 40 transactions after the shock. It shows clearly that as the size of the shocks increase, the IRFs tend to be more asymmetric between positive and negative shocks. The decrease in expected durations from a negative 90% shock is almost twice the size of the corresponding increase in expected duration from a positive 90% shock. For a linear ACD model, the responses are symmetric to shocks with same magnitude but different sign.

Investor's asymmetric responses to the signal of multiple transactions are also important features of the nonlinear dynamics of the estimated TACD for each subperiod. The parameter estimates in Table 4 show that the occurrence of multiple transactions will reduce the conditional mean durations and therefore increase the transaction intensity. The sizes of reduction are different across regimes. $\omega^{(2)}$ is significantly larger than $\omega^{(1)}$ in absolute value for each

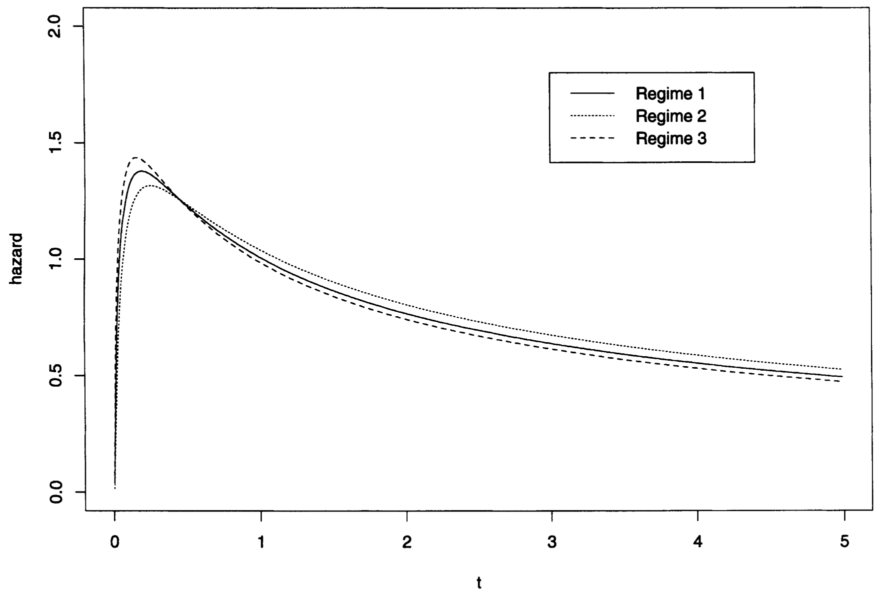

Fig. 4. Hazard functions for each of the 3 regimes of Period 2.

of the seven subperiods, but the relations between $\omega^{(2)}$ and $\omega^{(3)}$ are mixed. It is clear that the occurrence of multiple transactions has a nonuniform effect across regimes.

Hazard function, or conditional intensity function of a trading process, gives a complete characterization of the dynamics of trade arrivals given past information. We find that hazard functions are also different across regimes. We plot the hazard functions for Period 2 in Fig. 4. It shows the shape of the hazard function is of an inverted U shape for each regime as opposed to the monotonic shape allowed by Weibull specification. Hence, the transaction arrival intensity is low for very short and very long durations as it is first monotonically increasing and then gradually decreasing after reaching the peak.

6. Conclusions

We proposed a threshold ACD model for modeling transaction durations and provided conditions for geometric ergodicity and existence of moments of the model. The TACD model captured the nonlinear relation between the conditional expected duration and lagged durations as shown by the nonlinear test statistics. Relating economic variables suggested by market microstructure theories to the current regime enables us to identify the fast transacting regime as the informed trading regime and the slow transacting regime as noninformed trading regime.

Nonstationarity is found to be an important feature of the transaction duration data. Even with our flexible, nonlinear model we find evidence of structural breaks in the transaction duration data of IBM, and each break point can be aligned with real economic events. The estimated TACD model for each subperiod also revealed the changing dynamics through time periods.

In light of the joint modeling framework of Engle (2000), the proposed TACD model provides a better understanding of the intertemporal correlated waiting times between transactions, and it is an important step toward modeling economic variables observed at each transaction and the trading process as a whole.

Finally, systematic modeling procedures and complete characterizations of TACD models are needed. It is important to study the distributional theory of test statistics for TACD models and the impacts of the delay parameter of the model on the nonlinear dynamics. Joint estimation and testing of TACD and structural breaks are also of interest, as discussed in Koop and Potter (1997) for TAR and structural breaks.

Acknowledgements

The authors would like to thank the Co-Editor, Peter Robinson, two anonymous referees, Tim Bollerslev, K.S. Chan, Rong Chen, Robert Engle, Hong Li, Per Mykland, George C. Tiao and Pietro Veronesi for their suggestions.

Appendix A.

A.1. Proof of Theorem 1 (ergodicity of TACD(1,1) process)

Proof. We first write the process in a Markovian representation using state space approach and then show that the Markov chain is a φ-irreducible T-chain. Lastly, we use the drift criteria to show the geometric ergodicity of the process.

Markovian representation. Let $Y_i = \psi_{i+1}$, then the process can be written as a two dimensional Markov chain

$$\begin{cases} X_i = Y_{i-1}\varepsilon_i^{(j)} \\ Y_i = \sum_{h=1}^{J}[\alpha_0^{(h)} + (\alpha_1^{(h)}\varepsilon_i^{(j)} + \beta_1^{(h)})Y_{i-1}]\mathbf{1}_{\{Y_{i-1}\varepsilon_i^{(j)} \in R_h\}} \end{cases} \quad \text{if } X_{i-1} \in R_j. \tag{15}$$

The state space of the process is given by $S=\bigcup_{j=1}^{J} S_j$, where S_j is the closure of the domain of the chain for each regime and is given by $S_j=\{(x,y)\colon x\in [r_{j-1},r_j],\ y\geqslant \alpha_0^{(j)}+\alpha_1^{(j)}x+\beta_1^{(j)}\bar{y}\}$ and $\bar{y}=\min_{\beta_1^{(j)}<1}\{(\alpha_0^{(j)}+\alpha_1^{(j)}r_{j-1})/1-\beta_1^{(j)}\}$. Since $\beta_1^{(J)}<1$, $\bar{y}$ is well defined. The state space S is equipped with a σ-field $\mathscr{B}(S)$, generated with the Borel σ-field on $\mathbb{R}^2$ restricted to S.

φ-Irreducible T-chain. Let μ^{Leb} be the Lebesgue measure on S. We shall show that the Markov chain is μ^{Leb}-irreducible. For any two points, (x_0,y_0) in S and (x,y) in the interior of S, assuming $x_0\in R_k$, $x\in R_j$ and $\bar{y}$ is achieved in regime h. Let $y^*=(y-\alpha_0^{(j)}-\alpha_1^{(j)}x)/\beta_1^{(j)}$. Since $y>\bar{y}$, we have $y^*>\bar{y}$. Let $\delta=\min\{y^*-\bar{y},\alpha_1^{(h)}(r_h-r_{h-1})/(1-\beta_1^{h})\}$. Thus, there exists a positive integer $m=\min\{n\geqslant 1\colon \bar{y}+\delta/2+(y^*-\bar{y}-\delta/2)[\beta_1^{(h)}]^{-n}\geqslant \alpha_0^{(J)}+\alpha_1^{(J)}r_{J-1}+\beta_1^{(J)}y_0\}$, such that the point (x,y) can be reached from (x_0,y_0) through the following $m+2$ steps

$$(x_0,y_0)\to(x_1,y_1)\to(x_2,y_2)\cdots(x_m,y_m)\to(x_{m+1},y^*)\to(x,y),$$

where $y_i=\bar{y}+\delta/2+(y^*-\bar{y}-\delta/2)[\beta_1^{(h)}]^{i-(m+1)}$, $x_1=(y_1-\alpha_0^{(J)}-\beta_1^{(J)}y_0)/\alpha_1^{(J)}$ and $x_l=r_{h-1}+(\delta/2)(1-\beta_1^{(h)})/\alpha_1^{(h)}$ for $l>1$. Thus, the $m+2$ step transition density from (x_0,y_0) to (x,y) is positive since

$$\begin{aligned} p^{m+2}((x_0,y_0),(x,y)) \geqslant\ & f^{(k)}\left(\frac{x_1}{y_0}\right) f^{(J)}\left(\frac{x_2}{y_1}\right) f^{(h)}\left(\frac{x_3}{y_2}\right)\cdots \\ & f^{(h)}\left(\frac{x_{m+1}}{y_m}\right) f^{(h)}\left(\frac{x}{y^*}\right)>0. \end{aligned} \tag{16}$$

For any set $A\in\mathscr{B}(S)$ with $\mu^{\mathrm{Leb}}(A)>0$, there exists an open ball $B_\eta((x,y))\in R_j$ for some $1\leqslant j\leqslant J$ and a small positive number η, such that all the points in $B_\eta((x,y))$ can be reached from (x_0,y_0) in $m+2$ steps as in our above construction. Thus, we have

$$\sum_{n=1}^{\infty} P^n((x_0,y_0),A)\geqslant P^{m+2}((x_0,y_0),B_\eta((x,y)))>0. \tag{17}$$

By definition, the Markov chain $\{X_i,Y_i\}$ is μ^{Leb}-irreducible.

The one step transition probability of the Markov chain can be written as

$$P((x_0,y_0),A)=\int\!\!\int_{A(x_0,y_0)} \sum_{j=1}^{J} \mathbf{1}_{\{x_0\in R_j\}} f^{(j)}\left(\frac{x}{y_0}\right)\,\mathrm{d}x\,\mathrm{d}y, \tag{18}$$

where $A(x_0,y_0)=\{(x,y)\in A\colon y=\sum_{h=1}^{J}[\alpha_0^{(h)}+\alpha_1^{(h)}x+\beta_1^{(h)}y_0]\mathbf{1}_{\{x\in R_h\}}\}$. Using the same argument as in Proposition 6.3.6 of Meyn and Tweedie (1993), it can be shown that our process is a T-chain. Thus the Markov chain $\{X_i,Y_i\}$ is a φ-irreducible T-chain. This implies that all compact sets in S are petite and can be served as test sets.

Geometric ergodicity. We choose the test function of the form

$$V(x,y)=1+\tfrac{1}{2}[1-(\alpha_1^{(J)}+\beta_1^{(J)})]x+y \tag{19}$$

and the compact set of the form $C(M)=\{(x,y)\in S\colon x+y\leqslant M\}$, where M is a positive constant to be determined later.

Define $f(x,y):=\mathbb{E}[Y_{i+1}|X_i=x,Y_i=y]$. It can be expressed as

$$\begin{aligned} f(x,y) = \sum_{j=1}^{J}\Bigg\{&\sum_{h=1}^{J}\alpha_0^{(h)}\Pr(y\varepsilon_i^{(j)}\in R_h)\\ &+y\sum_{h=1}^{J}\mathbb{E}[(\alpha_1^{(h)}\varepsilon_i^{(j)}+\beta_1^{(h)})\mathbf{1}_{\{y\varepsilon_i^{(j)}\in R_h\}}]\Bigg\}\mathbf{1}_{\{x\in R_j\}}. \end{aligned} \tag{20}$$

Since

$$\lim_{y\to\infty}\sum_{h=1}^{J}\mathbb{E}[(\alpha_1^{(h)}\varepsilon_i^{(j)}+\beta_1^{(h)})\mathbf{1}_{\{y\varepsilon_i^{(j)}\in R_h\}}]=\alpha_1^{(J)}+\beta_1^{(J)},\quad \forall j=1,\ldots,J$$

thus, $\exists M_0>\bar{y}$, if $y>M_0$

$$f(x,y)<\eta_0 y-1\quad\text{where } \alpha_1^{(J)}+\beta_1^{(J)}<\eta_0<1-\frac{1}{2}[1-(\alpha_1^{(J)}+\beta_1^{(J)})].$$

Also noted that

$$f(x,y)\leqslant\max_{1\leqslant h\leqslant J}\{\alpha_0^{(h)}\}+\left[\max_{1\leqslant h\leqslant J}\{\alpha_1^{(h)}\}+\max_{1\leqslant h\leqslant J}\{\beta_1^{(h)}\}\right]y,\quad\forall(x,y)\in S.$$

Thus, the conditional expectation of the test function satisfies

$$\begin{aligned}\mathbb{E}[V(X_{i+1},Y_{i+1})|X_i=x,Y_i=y] &= 1+\tfrac{1}{2}[1-(\alpha_1^{(J)}+\beta_1^{(J)})]y\\ &\quad+f(x,y)<K_1+K_2y.\end{aligned}$$

Let $\eta=\eta_0+\frac{1}{2}[1-(\alpha_1^{(J)}+\beta_1^{(J)})]$ and $M=(2/\eta)(K_1+K_2M_0)/[1-(\alpha_1^{(J)}+\beta_1^{(J)})]+M_0$, then if $x+y>M$, we have

1. If $y>M_0$, then $\mathbb{E}[V(X_{i+1},Y_{i+1})|X_i=x,Y_i=y]<\eta y<\eta V(x,y)$.
2. If $y\leqslant M_0$, then $\frac{1}{2}\eta[1-(\alpha_1^{(J)}+\beta_1^{(J)})]x>(K_1+K_2M_0)$, thus,

$$\mathbb{E}[V(X_{i+1},Y_{i+1})|X_i=x,Y_i=y]<K_1+K_2M_0<\eta V(x,y).$$

Thus, based on Theorem 15.0.1 of Meyn and Tweedie (1993), the Markov chain $\{X_i,Y_i\}$ is geometrically ergodic, so is $\{X_i\}$. This finishes the proof. □

A.2. Proof of Theorem 2 (Existence of moments for TACD(1,1) process)

Proof. Let $c_1=\max_{1\leqslant j\leqslant J}\{\mathbb{E}[(\alpha_1^{(J)}\varepsilon_i^{(j)}+\beta_1^{(J)})^k]\}$ and $c_2=\max_{1\leqslant j\leqslant J}\{\mathbb{E}[(\varepsilon_i^{(j)})^k]\}$. By choosing the test function of the form: $V(x,y)=1+[(1-c_1)/(2c_2)]x^k+y^k$ and petite set of the form $C(M)=\{(x,y)\in S\colon x+y\leqslant M\}$. It is easy to check that the following two key facts hold:

1. There exist finite positive constants $\{b_l\colon l=1,\ldots,k\}$, independent of y, such that,

$$\begin{aligned}&\mathbb{E}[V(X_{i+1},Y_{i+1})|X_i=x,Y_i=y]\\&\leqslant 1+\left\{\frac{1}{2}(1-c_1)+\sum_{h=1}^{J}\mathbb{E}[(\alpha_1^{(h)}\varepsilon_i^{(j)}+\beta_1^{(h)})^k\mathbf{1}_{\{y\varepsilon_i^{(j)}\in R_h\}}]\right.\\&\qquad\left.+\sum_{l=1}^{k}b_l y^{-l}\right\}y^k\end{aligned}$$

2. $\lim_{y\to\infty}\sum_{h=1}^{J}\mathbb{E}[(\alpha_1^{(h)}\varepsilon_i^{(j)}+\beta_1^{(h)})^k\mathbf{1}_{\{y\varepsilon_i^{(j)}\in R_h\}}]=\mathbb{E}[(\alpha_1^{(J)}\varepsilon_i^{(j)}+\beta_1^{(J)})^k]\leqslant c_1\ \forall j.$

Then, as in the proof of Theorem 1, we can prove that there exist $\eta\in(0,1)$ and $M>0$, such that $\mathbb{E}[V(X_{i+1},Y_{i+1})|X_i=x,Y_i=y]<\eta V(x,y)$, $\forall(x,y)$ outside $C(M)$. Finally, the result follows from Theorem 14.0.1 of Meyn and Tweedie (1993). □

References

Andersen, T.G., Bollerslev, T., 1998. Answering the skeptics: Yes, standard volatility models do provide accurate forecasts. International Economic Review 39, 885–905.

Andrews, D.W.K., 1993. Tests for parametric instability and structural change with unknown change point. Econometrica 61 (4), 821–856.

Andrews, D.W.K., Ploberger, W., 1994. Optimal tests when a nuisance parameters is present only under the alternative. Econometrica 62 (6), 1383–1414.

Bai, X., Russell, J.R., Tiao, G.C., 1999. Beyond Merton's Utopia; effects of non-normality and dependence on the precision of variance estimates using high-frequency financial data. Working Paper, Graduate School of Business, University of Chicago.

Berndt, E., Hall, B., Hall, R., Hausman, J., 1974. Estimation and inference in nonlinear structural models. Annals of Economic and Social Measurement 3, 653–665.

Bollerslev, T., 1986. Generalized autoregressive conditional heteroskedasticity. Journal of Econometrics 31, 307–327.

Bougerol, P., Picard, N., 1992. Stationarity of GARCH processes and of some nonnegative time series. Journal of Econometrics 52, 115–127.

Brockwell, P.J., Liu, J., Tweedie, R.L., 1992. On the existence of stationary threshold autoregressive moving-average processes. Journal of Time Series Analysis 13 (2), 95–107.

Chan, K.S., Petruccelli, J.D., Tong, H., Woolford, S.W., 1985. A multiple-threshold AR(1) model. Journal of Applied Probability 22, 267–279.

Chen, R., Tsay, R.S., 1991. On the ergodocity of TAR(1) processes. Annals of Applied Probability 1 (4), 613–634.

Easley, D., O'Hara, M., 1987. Price, trade size, and information in securities markets. Journal of Financial Economics 19 (1), 69–90.

Easley, D., O'Hara, M., 1992a. Adverse selection and large trade volume. Journal of Financial and Quantitative Analysis 27 (2), 185–208.

Easley, D., O'Hara, M., 1992b. Time and the process of security price adjustment. Journal of Finance 47, 577–606.

Engle, R.F., 1982. Autoregressive conditional heteroscedasticity with estimates of the variance of United Kingdom inflation. Econometrica 50 (4), 987–1006.

Engle, R.F., 2000. The econometrics of ultra-high frequency data. Econometrica 68 (1), 1–22.
Engle, R.F., Russell, J.R., 1995. Autoregressive conditional duration: A new model for irregular spaced data. Unpublished manuscript, University of California, San Diego.
Engle, R.F., Russell, J.R., 1998. Autoregressive conditional duration: A new model for irregular spaced transaction data. Econometrica 66 (5), 1127–1162.
Friedman, J.H., 1984. A variable span smoother. Technical Report No. 5, Laboratory for Computational Statistics, Department of Statistics, Stanford University.
Glosten, L.R., Jagannathan, R., Runkle, D.E., 1993. On the relation between the expected value and volatility of the nominal excess return on stocks. Journal of Finance 48, 1779–1801.
Glosten, L.R., Milgrom, P.R., 1985. Bid, ask and transaction prices in a specialist market with heterogeneously informed traders. Journal of Financial Economics 14, 71–100.
Goodhart, C.A.E., O'Hara, M., 1997. High frequency data in financial markets: Issues and applications. Journal of Empirical Finance 4, 73–114.
Grammig, J., Maurer, K.-O., 2000. Non-monotonic hazard functions and the autoregressive conditional duration model. Econometric Journal 3(1), 16–38.
Koop, G., Potter, S., 1997. Nonlinearity, structural breaks or outliers in economic time series. Working Paper, University of Edinburgh.
Lancaster, T., 1990. The Econometric Analysis of Transition Data. Cambridge University Press, Cambridge.
Lee, C.M.C., Mucklow, B., Ready, M.J., 1993. Spreads, depths and the impact of earnings information: an intraday analysis. Review of Financial Studies 6 (2), 345–374.
Lunde, A., 1999. A generalized Gamma autoregressive conditional duration model. Working Paper, Department of Economics, Politics and Public Administration, Aalborg University, Denmark.
Meyn, S., Tweedie, R.L., 1993. Markov Chains and Stochastic Stability. Springer, London.
Moeanaddin, R., Tong, H., 1990. Numerical evaluation of distributions in non-linear autoregression. Journal of Time Series Analysis 11 (1), 33–48.
Olsen and Associates, 1995. First International Conference on High Frequency Data in Finance (HFDF-I). Olsen and Associates, Zurich, Switzerland.
Olsen and Associates, 1998. Second International Conference on High Frequency Data in Finance (HFDF-II). Olsen and Associates, Zurich, Switzerland.
Petruccelli, J., Woolford, S.W., 1984. A threshold AR(1) model. Journal of Applied Probability 21, 270–286.
Rabemananjara, R., Zakoian, J.M., 1993. Threshod ARCH models and asymmetries in volatility. Journal of Applied Econometrics 8, 31–49.
Tong, H., 1990. Non-linear Time Series: A Dynamical System Approach. Oxford University Press, Oxford, UK.
Tsay, R.S., 1989. Testing and modeling threshold autoregressive processes. Journal of American Statistical Association 84 (405), 231–240.
Tsay, R.S., 1992. Model checking via parametric bootstraps in time series analysis. Applied Statistics 41 (1), 1–15.
Tsay, R.S., 1998. Testing and modeling multivariate threshold models. Journal of American Statistical Association 93 (442), 478–493.
Zakoian, J.-M., 1994. Threshold heteroskedastic models. Journal of Economic Dynamics and Control 18, 931–955.